- Caspian sea and the black sea as the water boundaries between Europe and Asia.
- Asia and Oceania have a land boundary in the new guinea islands.
- Mount Everest(located between Nepal and China) is the highest point on the earth in the Himalayas.
- Dead Sea(Between Israel and Jordan) is the lowest point on the land surface.

The topic for the next class: Geomorphology

Geography Class 08

Geomorphology (5:09 PM)

- It is the study of the physical features of the earth and the process by which those features are formed.
- ***Origin of the earth:*** Earth originated from a gaseous nebula.
- As the nebula cooled rings of matter ejected resulting in planets.
- ***Evolution of earth:*** The planet earth was initially barren, rocky, and in a volatile state.
- This primordial earth gradually evolved into the present stable one with a thick atmosphere.
- ***Formation of interior layers of the earth:***
- Due to a gradual cooling down of the earth, the heavier elements started to sink toward the centre and the lighter ones moved toward to surface through the process of density separation.
- With the gradual increase in density at the centre, the interior temperature increased
- With time as the earth cooled further, it condensed into a smaller size(condensation).
- The further process of differentiation led to the formation of different layers in the earth's interior.
- ***Evolution of atmosphere and hydrosphere:***
- The early thin atmosphere with hydrogen and helium was stripped off due to solar winds.
- During the cooling of the earth, gasses and water vapour were released from the earth's interior.
- And the process through which these gases outpour into the atmosphere is called Degassing.
- The volcanic eruption released more gasses and water vapour.
- The important gases at this stage were carbon dioxide, nitrogen, methane, water vapour, and every little oxygen.
- As the earth cooled further the water vapour started to condense and the CO_2 in the atmosphere got dissolved in rainwater and was brought down to the surface.
- This further reduced the temperature of the atmosphere which led to more condensation and more precipitation.
- The rainfall from the atmosphere got collected in the depressions on the surface forming oceans which were completed around 4000 million years ago.
- Life evolved in the oceans 3800 million years ago as Non-photosynthesis micro-organisms.
- Between 3000-2000 million years ago, blue-green algae emerged in ocean water which released oxygen through photosynthesis.

- By 2000 million years ago ocean was saturated with oxygen and oxygen started flooding the atmosphere.

Geological time scale (6:13 PM)

- ***Proterozoic Eon:*** Soft body marine organism.
- ***Hadean Eon:*** Ocean and continent were forming.
- ***Archean Eon:*** Blue-green algae evolved.
- ***Phanerozoic Eon:*** It is divided into 3 different eras:
- **Palaeozoic era**: It means old life.
- It includes periods Cambrian(no terrestrial life), Ordovician(1st fish), Silurian(1st plant), Devonian(Amphibians), Carboniferous(286-360 Million years)(1st reptile), and Permian(domination of reptiles).
- **Mesozoic era**: It means medium life.
- This era includes the Triassic, Jurassic (208-144 million years)(Age of dinosaurs), and Cretaceous Periods(extinction of dinosaurs).
- **Cainozoic era:** The Cainozoic era is divided into periods of tertiary(evolution of mammals, Himalayas, and apes) and Quaternary(the period of homo-sapiens).
- We are currently in the Holocene epoch (0-11700 years).

Holocene epoch (7:13 PM)

- ***Ages of Holocene:*** Greenlandian age:
- The 1st age is called Greenlandian (8200-11700 years ago)
- **Northgrippean age:** 8200-4200 years ago is called as Northgrippean age.
- **Meghalayan age**: 4200 years-Present is the Meghalayan age.
- In Mawluh cave in Meghalaya, evidence of change of age was found thus the name of Meghalayan age.

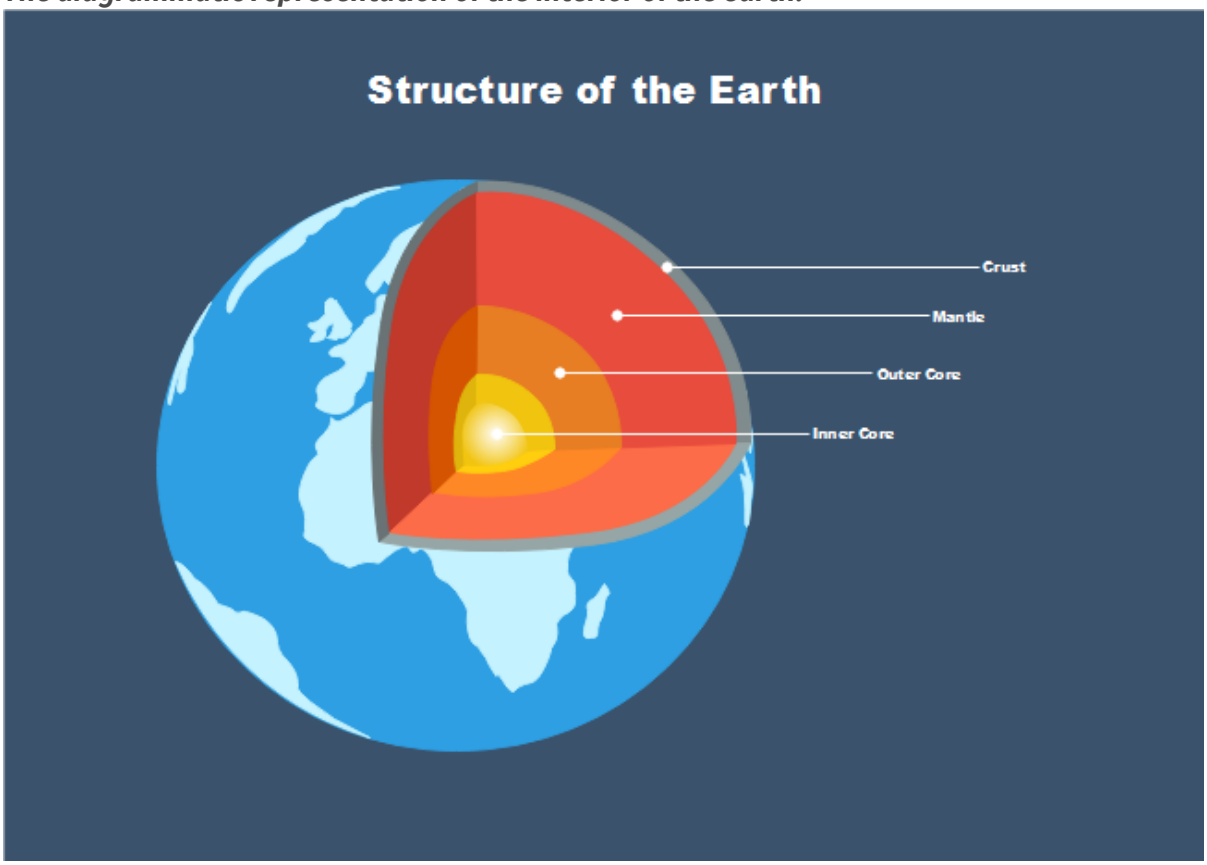
The interior of the earth (7:28 PM)

- ***The sources of study of earth's interior:***
- There are direct sources and indirect sources.
- ***Direct sources:***
- Mining and Deep ocean drilling(the maximum depth up to which we were able to drill is 12 km).
- Volcanic eruptions through which we can analyze the material from the earth's interior.
- ***Indirect sources:***
- *Density studies:*
- It is done by analyzing the average density of the earth(5.5 gm/cm3) and its comparison to the density of the surface(2.7-3 gm/cm3) and the core.
- we can conclude that the crust is lighter and the core is heavier.
- *Seismic study:*
- It is done by analyzing different earthquake waves, their speed, and their direction while passing through the earth's interior.
- *Tempratue and pressure:*
- The temperature increases by 1 degree celsius for every 32 meters near the surface.
- With the increase in depth pressure increases and the melting point of rocks increases.
- Using this correlation we can conclude about the earth's interior.
- *Meteorite:* It is by analyzing the structure, and mineralogy of meteorites we can conclude about the earth's interior as meteorites are remanet of planets.

The topic of the next class: The layers of the Interior of the earth.

Geography Class 09

Last class revision (5:09 PM)
Interior of the earth (5:20 PM)

- ***The diagrammatic representation of the interior of the earth:***
- 

- ***The chemical divisions of the earth's interior:***
- The 3 layers of crust, mantle, and core are the chemical division of the earth.
- ***Crust:***
- It is the uppermost layer of the earth.
- It is made up of light elements such as silica, aluminium, magnesium, etc.
- It is the lightest in terms of density.
- It is of 2 types, continental crust, and oceanic crust.
- ***The difference between the continental crust and oceanic crust:***
-

Continental crust:	**Oceanic crust:**
Very thick(35-45 km)	Very thin(8-10 km)
Lighter in density.	Heavier in density.
Rocks of brighter colour.	Rocks of darker colour.
Rocks are older	Rocks are younger

It is richer in Alumium, sodium, and Potassium elements(SiAl layer)	It is richer in Magnesium, calcium, and Iron. (SiMa layer)

- ***Mantle:***
- It is lighter than the core and denser than the crust.
- It accounts for 83% volume of the earth.
- It consists of 68% of the mass of the earth.
- The overall depth of the mantle extends to 2888 km.
- The mantle contains a higher proportion of magnesium and a lower proportion of silica and aluminium.
- It is divided into the upper and lower mantle.
- ***Core:***
- It is the innermost and densest layer of the earth.
- It is rich in nickel and iron therefore it is also called as NiFe layer.
- It is divided into the inner core and outer core.
- The inner core is solid and the outer core is liquid as the pressure at the inner core is very high.

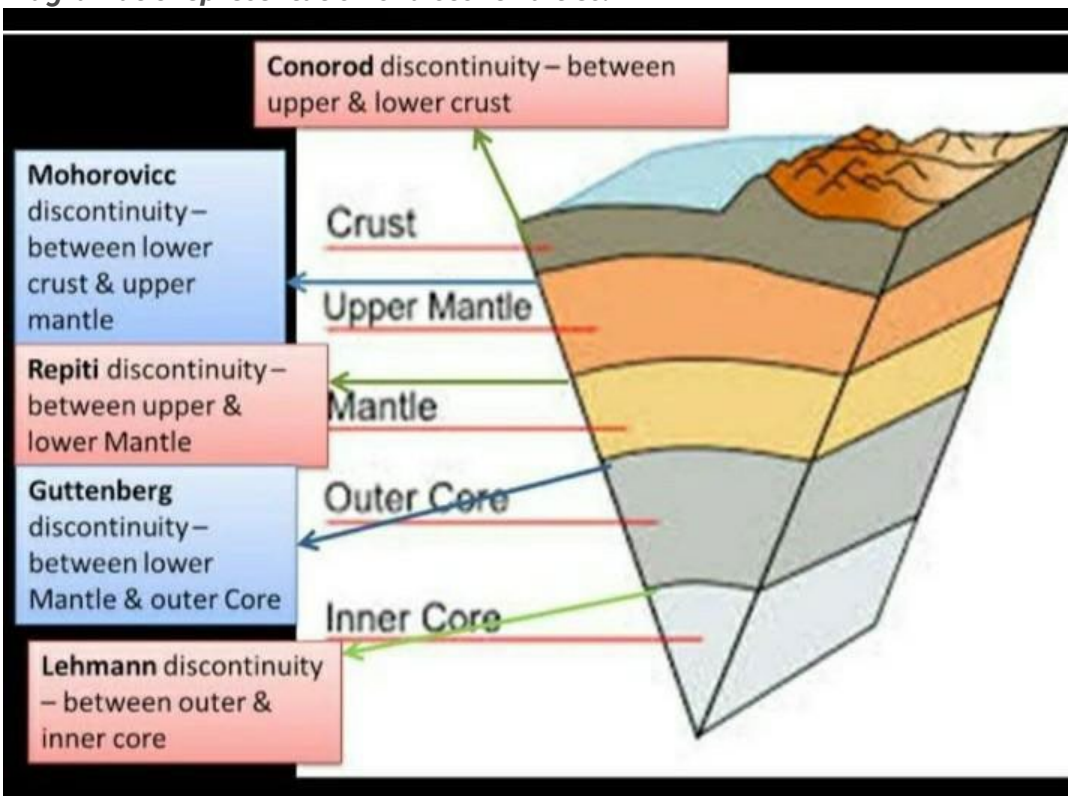
Physical division of the interior of the earth (5:58 PM)

- ***Lithosphere:***
- It is a solid layer made up of Crust and the upper part of the Upper Mantle.
- It is a hard and rigid outer layer that is divided into different plates.
- The thickness of the lithosphere is nearly 100 Km.
- ***Asthenosphere:***
- It extends between 100-400 km.
- It is in a semi-solid and semi-liquid state and undergoes deformation under pressure.
- It is also a source of magma on the surface.
- It is also called a low-velocity zone due to the slowing down of earthquake waves in this zone.
- ***Mesosphere***
- It includes the rest of the Mantle.
- ***Barrysphere:***
- It includes both the outer and inner core.

Composition of the earth (6:17 PM)

- ***Elements of the entire earth:***
- Iron: 35%
- oxygen: 30%
- Silicon- 15%
- Magnesium: 13%
- ***Major Elements of the Earth's Crust:***
- Oxygen: 46.60%
- Silicon: 27.72%
- Aluminium: 8.13%
- Iron: 5%.

Discontinuities in the interior of the earth (6:21 PM)

- It is the transition zone between the different layers of the earth's interior with different physical and chemical characteristics.
- ***Diagramatic representation of discontinuities:***
- 

- There are mainly 5 types of discontinuities:
- ***Conrad Discontinuity:***
- It is Within the Crust (Upper and Lower Crust).
- ***Mohorovicic or Moho Discontinuity:***
- It is Between Crust and the Mantle.
- ***Repetti Discontinuity*:**
- It is Between the Upper and the Lower Mantle.
- ***Guttenberg Discontinuity:***
- It is Between Mantle and Outer Core.
- ***Lehmann Discontinuity:***
- It is Between the Outer and Inner Core.

<u>Types of rocks (6:32 PM):</u>

- Any naturally occurring aggregate of minerals is called a rock.
- ***Difference Between Rocks and Minerals:***
-

Rocks:	Minerals:
Rocks are an aggregate of mineral elements.	Minerals are solid inorganic substances occurring naturally.
A rock has no definite chemical composition	Minerals have a definite chemical composition.

Minerals are organized to form rocks.	Elements are organized to form compounds that are known as minerals. An element is a pure substance.
The three chief types of rocks are- Igneous, Sedimentary, and metamorphic.	Four chief mineral groups are- Silicates, Carbonates, Sulphides, and metallic minerals.
Basalt, Granite, Sandstone, Slate, and Quartz are some important types of rocks.	Iron, Silicon, Magnesium, Nickel, and Calcium, are abundant minerals of the earth.

Types of rocks (7:02 PM)

Igneous rocks:

- They are formed due to the cooling, solidification, and crystallization of molten material from the earth's interior.
- They are also called primary rocks.
- Igneous rocks are of two types, intrusive and extrusive igneous rocks.
- ***Intrusive igneous rocks:***
- They are formed from the cooling of magma inside the earth.
- The slow cooling of magma results in a crystalline texture with higher strength.
- They are also called Plutonic rocks.
- Examples are granite, Gabbro, etc.
- ***Extrusive igneous rocks:***
- They are formed due to the cooling of magma above the earth's surface.
- Faster cooling results glassy texture but with lower strength.
- They are also called Volcanic rocks.
- Examples are Basalt, Andesite, etc.
- Igneous rocks are also be classified as:
- ***Acidic or Silsic:***
- Where the silica percentage is more than 66%.
- For example Granite.
- ***Basic or Mafic:***
- Where the silica percentage is less than 52%.
- For example Basalt.

Sedimentary rock (7:29 PM)

- These are the secondary rocks formed by the solidification of sediments of original igneous or metamorphic rocks or another sedimentary rock.
- ***Stages in the formation of sedimentary rock:***
- Weathering>>Transportation(by a different agent such as wind, water, glacier, etc)>>Deposition(in a basin)>>Lithification(conversion of loose sediments into hard rock)>>Compaction(sediments are squeezed by the weight of overlying layers)>>Cementation(It involves binding together of compacted sediments by natural cementing material such as silt).

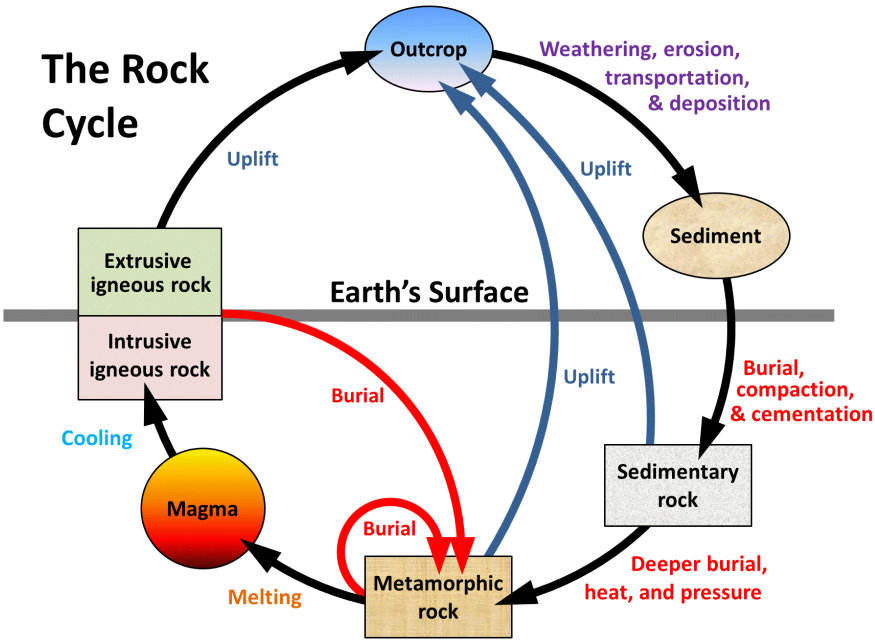
- Sedimentary rocks are formed in different layers or strata.
- They contain fossil evidence.
- Examples of sedimentary rocks are Sandstone, Limestone, Shale, Clay, Chalk, Coal, and Gypsum.

Metamorphic rock (7:47 PM)

- Metamorphism involves changes in the form of rocks through physical or chemical processes.
- The change in pressure conditions results in dynamic metamorphism and the change in temperature results in thermal metamorphism, together it causes thermo-dynamic metamorphism.
- During the process of Metamorphism if the minerals are arranged in a series of bands along a plain it is called foliation.
- foliation involves banding.
- When minerals are arranged linearly it is called Lineation.
- ***Examples of Metamorphics rocks:***
- Granite resulting in Gneiss.
- Limestone resulting in Marble.
- Sandstone resulting in Quartize.
- Shale resulting in Schist.
- Clay resulting in Slate.
- Coal results in Graphite and Diamonds.

The Rock Cycle (8:01 PM)

- ***The diagrammatic representation of the Rock cycle:***



-
- In nature, all the rocks get converted into each other form.
- The outcrop is the elevated part of the surface thus it undergoes weathering very easily.

The topic for the next class: Earth's Movements.

Geography Class 10

Last class revision (5:14 PM)

Earth's movement (5:30 PM)

- Various forces from the earth's interior as well as outside the earth's surface cause physical stress and chemical actions on the earth's material bringing about changes in the configuration of the surface of the earth called a geomorphic process.
- The geomorphic process results from two types of forces, endogenetic and exogenetic.
- ***Endogenetic forces:***
- These are the forces acting from the earth's interior, the source of energy is radioactivity and primordial heat.
- They result in large-scale upliftment or subsidence or folding and faulting.
- They are responsible for the formation of major structural units of the earth's surface.
- ***Exogenetic forces:***
- These are the forces acting on the earth's surface from above the surface.
- The sources of energy are sunlight and gravity.
- Wind, water, and glaciers are the different agents of exogenetic movements.
- These forces result in minor topographical features such as valleys and caves.

Endogentic movements (6:11 PM)

- They are classified into dystrophic and catastrophic movements.
- *Catastrophic movements (Sudden movements):*
- These are unpredictable movements of shorter duration.
- These movements can be observed.
- Examples are earthquakes, volcanic eruptions, etc.
- *Diastrophic movements:*
- Some movements move, elevate, and buildup up the portions of the earth's crust.
- These movements operate very slowly and led to the formation of primary landforms.
- They are of 3 types: tectonic, isostatic, and eustatic.
- *Tectonic movement:*
- The Word tectonic means to build.
- These are continent and mountain-building movements affecting the earth's surface.
- They are of 2 types Epirogenic and orogenic.
- *Epirogenic movements:*
- These are vertical movements caused by radial forces.
- These are characterized by large-scale upliftment/ emergency and subsidence/submergence of land areas.
- They are very slow and widespread.
- These movements led to the continental building.
- An example is the gradual upliftment of the Deccan plateau and the slow subsidence of the northern coast of the Gulf of Mexico.
- *Orogenic movement:*
- These are mountain-building movements caused by tangential forces.
- They involve intense folding and faulting of narrow belts.
- The tangential forces are of 2 types, Compressional resulting in folding, and tensional resulting in faulting.
- Folding:

- When the compressional forces the earth's rocks to push or squeeze against each other.
- It results in the formation of folds
- Folds are structures in which the layers are bent or distorted without the loss of continuity.
- Faulting is when tensional forces pull the rocks apart, it results in the formation of faults.
- Faults are the fractures, through which primary surfaces are broken and displaced, with the loss of continuity.
- ***Type of faults:***
- The up-folds are called anticlines and the Downfolds are called synclines.
- A fold will have two sides called limbs.
- Symmetrical Fold:
- If the limbs of a fold are inclined at the same angle, it is called a Symmetrical fold.
- Asymmetric Fold:
- If one of the limbs is inclined more than the other, it is called an Asymmetrical fold.
- Overfold:
- It is a fold where the strata in one limb have been folded beyond the vertical axis.
- Recumbent Fold:
  It is a fold lying down resulting from the continuation of pressure on one of the limbs.
- Nappe:
- It is formed when the pressure exerted upon a Recumbent Fold is sufficiently great to cause it to be torn from its roots and thrust forward.
- Nappes are well-developed in the Himalayas.
- ***Types of Faults:***
- Normal Fault:
- When one of the blocks moves downward relative to the other due to tensional forces it is called a Normal fault or Dip Slip Fault.
- Reverse Fault:
- When one block moves up about the other one due to compressional forces it is called a Reverse Fault or Thrust Fault.
- Strike-Slip Fault:
- Where both the blocks move across each other involving no vertical movements it is called Strike-slip or Transform Fault.
- In a fault, the upthrust block is called a Horst and the down-dropped block is called a Graben.

<u>The difference between the fold and the fault mountains(8.03 PM):</u>

-

Fold mountains:	**Fault/block mountains:**
It is because of the folding process.	Due to the Faulting process
Only compressional force	Both compressional and tension force

They are greater in length	They are greater in width.
For example Himalayas and Alps	For example Vindhyas and Blackforest

The topic for the next class: Isostatic movement and Eustatic movement.

Geography Class 11

Last class revision(5.10 PM).

Isostatcy/Isostatic movements(5.22 PM):

- Isostasy is the state of equilibrium or balance in the earth's crust.
- Isostatic movements involve vertical movements under the action of floatation displacement between the rock layers of differing density and mobility.
- This is to achieve balanced crustal columns of uniform mass above a level of compensation in which the topographic elevation is inversely related to underlying rock density.
- For example, the Mountains have deep roots, like the Scandinavian mountains due to the melting of ice sheets are observing the gradual rising of the land which is evident in a series of raised beaches.

Eustatic movements(5.44 PM):

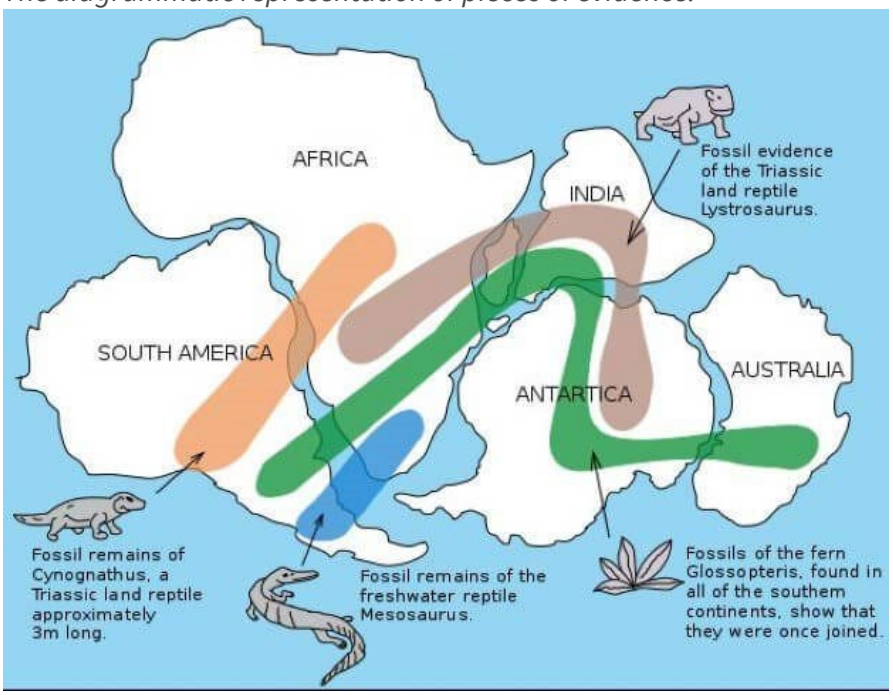
- They involve the worldwide movement of sea level resulting from changes in the total volume of liquid seawater or capacity of ocean basins.
- The volume of seawater can be changed by melting or the formation of glaciers.
- The capacity of the basin can be changed through the formation of ridges or the expansion of basins.

Continental drift theory(5.55 PM):

- The continental drift theory was proposed by Alfred Wegner(a german meteorologist) in 1912.
- The theory was proposed to explain major variations in the earth's climate.
- ***Assumptions*:**
- The three layers of the earth with outer SiAl, intermediate SiMa, and inner NiFe.
- The continental masses were assumed to be floating on oceanic crust without any resistance.
- ***The Theory:***
- Before the Carboniferous period(280-250 million years ago), there was only one supercontinent called Pangea and one superocean called Panthalassa.
- This Supercontinent started to rift during the Carboniferous period.
- It was split into northern Angaraland(Laurasia) and southern Gondwanaland by a rift running east to west.
- Gradually this rift enlarged to form the Tethys Sea.
- The Angaraland consisted of North America, Greenland, and Eurasia without India and Arabia.
- The southern Gondwanaland consisted of Africa, South America, India, Australia, and Antarctica.
- A North-South rift separated North America from Eurasia and South America from Africa which started to move towards the West.
- India started moving toward the North.
- Australia got separated from Antarctica and moved toward the northeast

- Africa moved towards the north.
- Finally, Arabia got separated from Africa and merged into Asia.
- ***Forces responsible for the continental drift:***
- Alfred Wegner proposed the following forces as the cause of continental motion:
- Equatorward or North-South movement is caused by the Pole-fleeing force due to Gravitational differential force and the force of Buoyancy.
- The westward movement is caused by the tidal forces of the Sun and Moon.

Evidence in support of continental drift theory(7.12 PM):

- *The diagrammatic representation of pieces of evidence:*
- 

- *Justafix or zig-saw fits of continents:*
- There are similarities in coastlines on opposite sides of the Oceans.
- All the continents can be merged to form one big continent.
- *Structural pieces of evidence:*
- The nature of physiography structure in different parts of the continents, having the same age and structural properties
- The mountain belts of Brazil terminate on the South American east coast and the same type of mountains reappeared again in Africa.
- *Stratigraphic pieces of evidence:*
- The eastern coast of brazil has the same type of rock formations observed along with Northwest Africa.
- *Fossil evidence:*
- Mosasaurus is an aquatic reptile whose fossil remains are found only in South America and South Africa separated by a wide ocean.
- The fossils of Glossopteeris(a fern) grown only in subpolar climates are now found in warm climatic regions separated by wide Oceans.
- *Glacial deposits:*
- The layers of tillites are found in warm tropical regions like South America, South Africa, Australia, and India.
- *Placer deposits:*

- The rich deposits of gold Placer near the Ghana coast without any source of Gold nearby.

Criticisms of continental drift theory(7.44):

- The forces suggested for the movement of the continent are considered to be inadequate.
- The rocks of continental crust and oceanic crust are very rigid and would not permit the drifting of continents over the oceanic floor.
- The theory did not describe the situations of pre-carboniferous times.

Mapping(7.49 PM):

- ***Africa:***
- It is also called the dark continent as it was discovered very late.
- It is home to the largest desert(the Sahara desert).
- It is home to the longest river on the earth that is the Nile river.
- The Nile is made of 2 rivers, the blue and white Nile.
- Mount Kilimanjaro is the highest point in Africa.
- The lowest point in Africa is the lake Assal(in Djibouti).

The topic for the next class: Seafloor spreading and plate tectonic theory.

Geography Class 12

Last class revision (5:10 PM)

Seafloor spreading (5:17 PM)

- *Mapping of the oceanic floor revealed the following information:*
- Presence of mid-oceanic ridges along the seafloor.
- The ridges are volcanically active resulting in a continuous eruption of magma.
- Rocks on either side of the ridges are of the same age with similar composition and magnetic properties.
- The age of rocks along the oceanic floors increases away from the ridge.
- The rocks of the oceanic crust are younger than the continental crust.
- The oceanic crust is thinner than the continental crust.
- *Theory:*
- Based on the above observation, Herry Hess proposed the theory of seafloor spreading in 1961.
- According to it, the continued magma eruption at the oceanic ridges causes a rupture of the oceanic crust.
- The new lava wedges on to oceanic crust, which pushes the oceanic crust onto either side therefore the ocean floor spreads.
- The spreading crust sinks at the oceanic trenches and gets consumed.

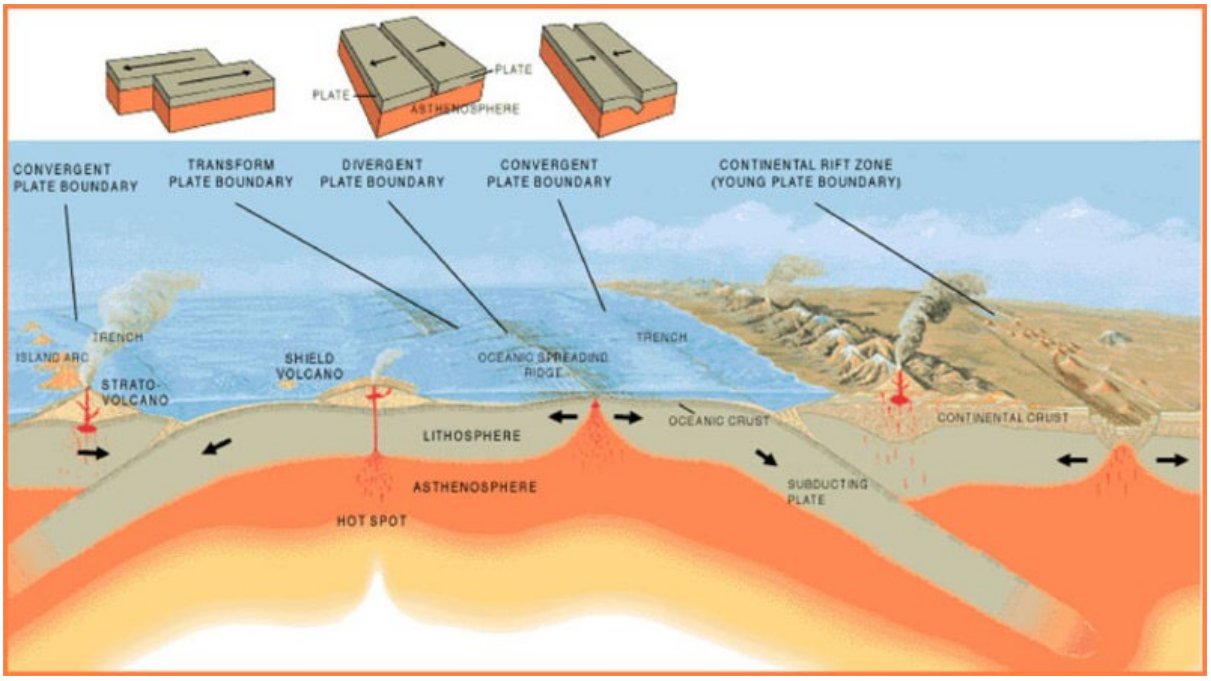
Plate tectonics theory (5:46 PM)

- *Introduction:*
- The term plate was coined by JT Wilson in 1965.
- The theory of plate tectonics was proposed by Morgan, Mckenzie, and Parker in 1967.
- *Plate tectonics:*
- Plates are broad and rigid segments of the lithosphere which includes the ridge upper part of the upper mantle and crust.
- The plates are in motion on underlined asthenosphere, which is in a semi-solid and semi-liquid state.

- Plate tectonics is the study of deformation within plates and of the interaction of plates around their margins.
- Plates are nearly 100 km thick and have high rigidity and are unable to deform except in response to very strong and prolonged force.
- There are 7 major plates on the earth's surface, pacific, north American, south American, Eurasian, Indo-Australian, and Antarctican.
- There are many minor plates such as Nazca Plates, cocos plates, Arabain plates, etc
- *The difference concerning continental drift theory:*
-

CDT	PTT
Sial, Sima, Nefi	Lithosphere(SiAl and Sima) and asthenosphere
Freely floating SiAl	Offer High resistance

- *Plate movements:*
- The plates are constantly in motion but with different speeds and directions, this causes 3 types of plate boundaries, Divergent, convergent and transform.
- A plate boundary or margin is a zone of motion between two plates.
- *Diagrammatic representation of different plates:*
-



Divergent/constructive plate boundary (6:32 PM)

- ***The ocean-ocean divergent plate boundary:***
- It is a type of plate boundary where two oceanic plate margins are moving apart in opposite directions
- It is a zone of tension where the lithosphere split and hot magma comes up through the cracks and solidifies leading to the formation of a new oceanic crust.

- The continues built of solid magma results in the formation of mid-oceanic ridges along the plate margins.
- In this plate boundary, shallow earthquakes with a focus of up to 70 km are observed.
- ***The continental-continental divergent plate boundary:***
- The formation of divergent plate boundaries along continents involves 3 stages:
- Intra-continental rifting:
- The upward movement of magma below the continental crust causes the fragmentation of the continent through the creation of numerous cracks and faults.
- Such a series of faults is called a rift valley.
- Rising magma starts to come out through this rift.
- For example, the East African rift valley.
- Inter-plate thinning:
- It involves partial melting of the lithosphere and gradual thinning of continental crust.
- Rift valley starts to widen and may gradually get filled with ocean water resulting in the formation of a shallow sea.
- For example, the Red sea.
- Formation of mid-oceanic ridge:
- The continues spreading of continental plates away from each other and the creation of new oceanic crust along the rift valley by the rising magma pulls the continental mass sufficiently apart.
- At this stage, there will be a new oceanic basin along both the sides of mid-oceanic ridge.
- For example mid-Atlantic ridge.

Convergent/destructive plate boundaries (7:42 PM)

- *Oceanic-oceanic convergence:*
- When two oceanic plate margins converge the ocean plate of higher density decent into the asthenosphere.
- This process is called subduction and the zones are called subduction zones.
- Subduction leads to the formation of trenches, that is the deepest regions on the surface of the earth.
- These subduction zones are the sites of the most widespread and intense earthquakes.

The topic for the next class: Convergent/destructive plate boundaries to continue.

Geography Class 13

Last class revision(5.13 PM).

Ocean-Ocean convergence(5.20 PM):

- When 2 oceanic plate margins converge oceanic plates of higher density descend into the asthenosphere, this process is called a subduction zone.
- The process of subduction leads to the formation of trenches which are the deepest regions on the surface of the earth.
- For example subduction of the pacific plate below the Eurasian plate has led to the formation of the Mariana Trench, and Japan Trench.
- The subducting plate undergoes deformation, intense compression, metamorphism, and melting as it reaches the deeper parts.

- Some of the molten material from the subducting plate rises upwards and accumulate continually on the other oceanic plates, this results in the formation of Volcanic mountains on the seafloor.
- When these Volcanic mountains rise above the sea level it results in the formation of volcanic islands.
- The continued volcanic activity gradually increases the size and elevation of volcanic islands.
- These islands are arranged parallel to the trenches in an arc shape and are called island arcs, for example, japan, Aleutian, and the Caribbean islands.
- An Archipelago is a group of scattered islands in the ocean formed due to ocean-ocean convergence plate boundary with intense volcanic activity, for example, Indonesia, Philippines, etc.
- A subduction zone is the site of widespread volcanic activity and earthquakes.
- In ocean-ocean convergent, all 3 kinds of earthquakes shallow, intermediate, and deep-seated are found.
- The majority of the earthquake appears to be confined to a dipping zone along the subduction plate called Benioff Zone.

Ocean-Continent convergent plate boundary(5.35 PM):

- When a plate with an oceanic margin collides with a plate of the continental margin oceanic crust is denser and subducts beneath the continental plate which is more buoyant, for example, the Nasca plate subducting below the South American plate.
- During the subduction, the thick sequence of rocks along the continental margin is compressed and deformed to form a chain of fold mountains, for example, Andes and rocky mountains.
- As the oceanic plate subducts it creates trenches and earthquakes are generated at Benioff zones.
- As the oceanic plate subducts deeper it gets melted and the magma starts to rise.
- The rising magma accumulates within the adjacent continental crust which eventually rises to the surface resulting in the formation of volcanoes.
- For example, the subduction of the Nasca plate below the South American plate has resulted in a series of volcanos in the Andes, such as Ozes del Salado and Cotopaxi.
- St. Helens volcano in the Rockies is also similar.
- Ocean-continent convergence results in all the 3 types of earthquakes.
- The Pacific Ocean is surrounded by trenches on all sides, which are zones of intense volcanic activities and earthquakes, therefore it is called as Pacific ring of fire.

Continental-continental convergence(6.09 PM):

- Before the collision of the continental margin, the land masses are usually separated by the oceanic crust.
- As the two plates converge, the oceanic crust located in between starts to subduct beneath one of the plates.
- The oceanic plate completely breaks from the continental block and assimilated into the mantle, at this point, Volcanic activity stops.
- The continued convergence causes the two continual masses to get stitched together along a zone called a suture zone.
- When continental margins continue to collide the low density of continental material doesn't permit its subduction.

- The continued convergence forces the continental crust partially under the other one creating an unusually thick layer of intense folding.
- The sediments deposited in the basis between the continental margins undergo continued compression resulting in the formation of very high-fold mountains, for example, the Himalayas.
- Shallow and intermediate earthquakes are absorbed along this margin.

Parallel plate/Conservative plate margin(6.39 PM):

- At the conservative plate margins, the plate slightly passes each other without any formation of new crust.
- The transform fault roughly moves parallel to the direction of plate movement, for example, the San Andreas fault of Califonia.
- A conservative plate margin involves no volcanic activity.
- The earthquakes observed are shallow in depth but they are more frequent and intense.
- Such a series of transform faults are also found along the oceanic plate margins where they offset the oceanic ridges throughout their length.

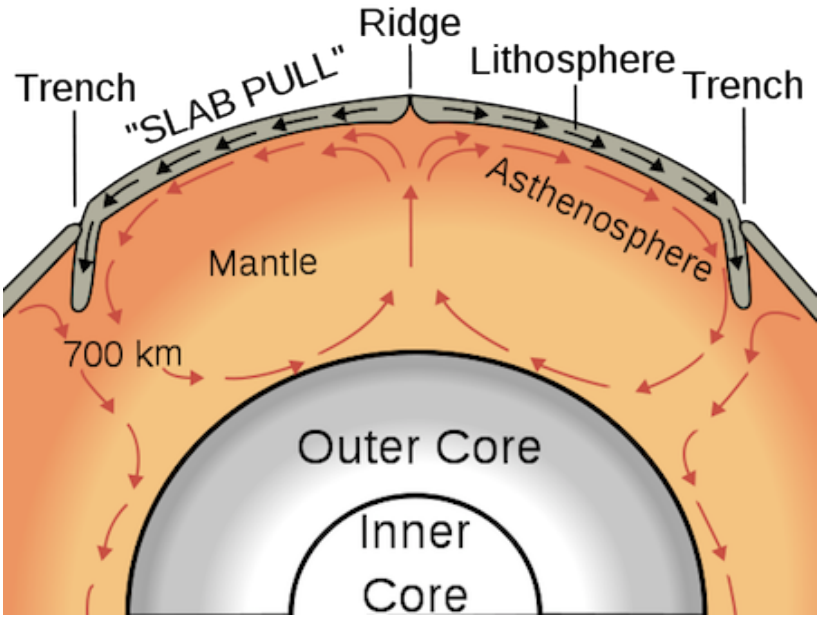
Table summarizing plate boundaries(7.16 PM):

-

	Divergent(O-O) plates	Divergent(C-C) plates	Convergent(O-O) plates	Convergent(O-C) plates	Convergent(C-C) plates
Features.	Mid-oceanic ridges, transform faults.	Rift valley, shallow sea, MOR.	Trenches, subduction zones, archipelago, island arcs, Benioff zone.	Trenches, subduction zones, fold mountains, volcanoes Benioff zone.	suture zone, partial subduction, geo-syncline, fold mountains, nappe.
Earthquakes.	shallow.	shallow.	shallow, intermediate, and deep.	shallow, intermediate, and deep.	shallow and intermediate
volcanism	Yes	Yes	Yes	Yes	No
Examples	Mid Atlantic Ridge, Carlsberg Ridge.	East African rift valley, Red sea, Mid Atlantic ridges	Japan, Aleutian, Indonesia, etc.	Andes, Rockies mountains.	The Himalayas.

Causes of Plates movements(7.32 PM):

- *Diagramatic representation of various causes:*

- 

- *The convection currents:*
- They are generated due to the intense heat released from the interior of the earth.
- The theory of convention current was proposed by Arthur homes.
- As the currents ascend from below they diverge and spread laterally.
- *Mantle plumes:*
- It is a type of mantle convention that involves jet-like plumes of low-density magma material from the core-mantle boundary.
- Mantle plumes are also the cause of the formation of volcanic hot spots, for example, reunion, Hawai, etc.
- *Ridge push:*
- The magma rushes along the oceanic ridges from wedges of the new lithosphere on either side trail edge of the plate and causes the plates to be pushed apart.
- *Gravity sliding:*
- The spreading centers along the mid-ocean ridges stand high on the oceanic floor this results in the gravitation side of the lithospheric slab being away from the oceanic ridges.
- *Slab pull:*
- It is considered to be along the subduction zone where the subducting plate pulls the rest of the slab along.

The topic for the next class: Dictation for causes of plate movement and volcanism.

Geography Class 14

Last class revision (5:11 PM)

Causes of Plates movements (5:28 PM)

- *The convection currents:*
- Arthur homes proposed the theory of convention current.
- There are two types of convection current, one involves the whole of the mantle and the second involves only the asthenosphere.
- They are generated due to the intense heat released from the interior of the earth due to radioactivity.
- As the currents rise from below, diverge and spread laterally.

- The convection causes the lithosphere to crack and rift forming a mid-oceanic ridge.
- As the plates move laterally the currents carry the slab of the lithosphere with them.
- When these currents encounter a similar current from the opposite direction, they descend into the deeper part of the mantle and drag the lithosphere along with them causing subduction.
- *Mantle plumes:*
- It is a type of mantle convention and involves jet-like plumes of low-density magma material from the core-mantle boundary.
- As the plume reaches the lithosphere it spreads out latterly doming the surface zones of the earth and moving them along the direction of the mantle plume.
- Mantle plumes are also the cause of the formation of volcanic hot spots, for example, the reunion hotspot, Hawai hotspot, etc.
- Mantle plumes through a supply of magma transfer heat to the lithospheric plates and keeps them in active motion.
- *Ridge push:*
- The magma rising along the oceanic ridges creates new lithosphere along either side of the plates.
- This causes both plates to be pushed apart.
- *Gravity sliding:*
- The spreading centres along the mid-ocean ridges stand high on the oceanic floor.
- This results in the gravitation sliding of the lithospheric slab away from the oceanic ridges.
- *Slab pull:*
- It is experienced along the subduction zone.
- As the subducting plate descends it pulls the rest of the slab along.

The criticism of plate tectonic theory (5:51 PM)

- The theory although widely expected is not able to explain the following:
- 1. Both Africa and Antarctica are surrounded by ridges but have no subduction zones.
- The future of these plates is not clear yet.
- 2. Plate tectonics in the past like the formation of earlier mountains, movements of different plates, and how they were different from the present ones.

Volcanism (6:08 PM)

- Volcanism includes all the phenomena associated with the movement of molten material from the interior of the earth to the surface.
- *It involves three stages:*
- 1. Generation of magma in the earth's interior that is asthenosphere and the mantle due to an increase in heat, decrease in pressure or increase in water content, which lowers the melting point of rock causing it to generate magma.
- 2. The intrusion of magma from lower layers into the lithospheric layers
- 3. Extrusion of magma when sufficient pressure builds up in lithospheric chambers the magma erupts onto the surface of the earth.
- *Magma and Lava:*
- Magma is hot, mobile and moltem silcate material.
- It is made up of a combination of solids, liquids, and gases.
- When magma reaches the surface, it erupts as lava.

- Lava that erupts is hotter and more volatile
- It cools either on the surface or underwater.
- *Types of Magma:*
- There are mainly two types of magma:
-

Granitic magma:	**Basaltic magma:**
It has a high silica content(more the 66%).	It has less than 52% silica content.
It has a lower density	It has a higher density
The Continental crust is made up of granite rocks	The oceanic crust is made up of basaltic rocks
The temperature is lower.	The temperature is higher.
Viscous.	Fluid.
It is generated at ocean-continent convergence and it is generated rarely.	It is generated at O-O divergence C-C divergence, hotspots, etc.

Magma generation and plate tectonics (7:03 PM)

-

	O-O Divergence	C-C Divergence	O-O Convergence	O-C Convergence	Hot spots
Location:	Mid oceanic Ridge	Rift Valley	Along the subduction Zone through volcanic mountains or islands.	Along the subduction Zone through volcanoes along the fold mountains	Intra-plate Hotspots
Cause:	Convection Currents	Convection current	Melting of Subducting plate.	Melting of Subducting plate.	Mantle Plume

Type of Magma:	Basaltic	Basaltic	Andesitic	Andesitic	Basaltic
Nature of eruption:	Fissure. Slow and continues.	Along volcanoes of the rift valley. Slightly explosive.	Violent and explosive	Violent and explosive	Smooth

Distribution of volcanoes (7:42 PM)

- *Ridge volcanism:*
- Along the oceanic floor through mid-oceanic ridges
- For example mid-oceanic ridges in the Atlantic Ocean, East Pacific Rise, and Carlsberg ridge.
- *Arc Volcanism:*
- Found along ocean-ocean convergence boundaries forming island arcs.
- For example Japan, the Aleutian Islands, the Caribbean islands, etc.
- *Volcanic chains:*
- Found along ocean-continent convergent plate boundaries with a straight chain of volcanoes on the continental margin.
- For example Andes and Rockies.
- *Volcanic clusters:*
- Found along the continent-continent divergent boundary.
- For Example east African rift valley
- *Volcanic lines:*
- Line of volcanoes arranged along with the hotspots.
- For Example Hawaii islands and the Reunion islands.

Volcanic landforms (7:50 PM)

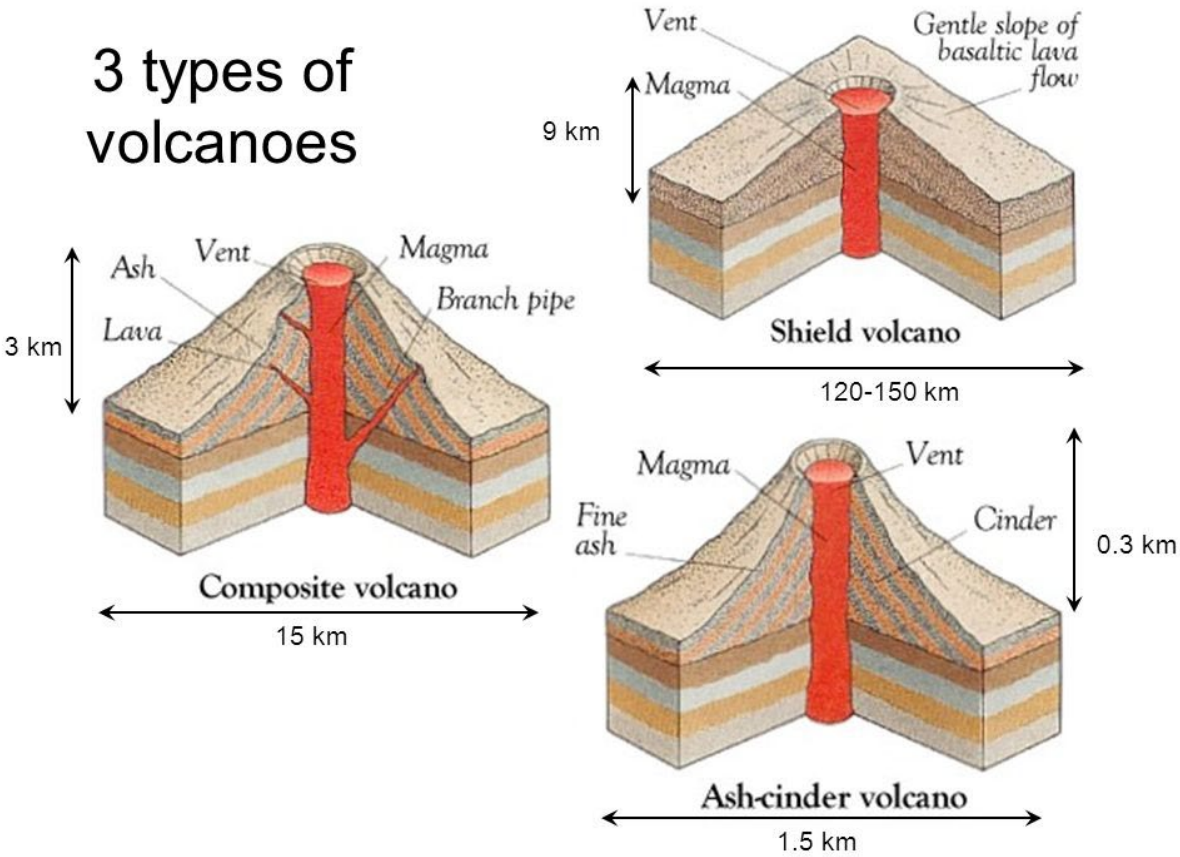
- There are two types of volcanic landforms, Intrusive and extrusive landforms.
- *Extrusive landforms:*
- Volcanic cones.
- Flood basalt province.
- Hot springs.
- Geysers.
- Fumarole.
- Mud volcano.
- *Intrusive landforms include:*
- Batholiths.
- Laccoliths.
- Lopoliths.
- Sill.
- Dykes.
- *Type of Volcanic cones:*
- There are 3 major types of volcanic cones:
- Shield volcano.
- Ash-cinder volcano.
- Composite volcano.

The topic of the next class: Volcanism (to continue)

Geography Class 15

<u>Last class revision (5.13 PM).</u>
<u>Extrusive Volcanic landforms (continued) (5.25 PM)</u>

- ***Volcanic cones:***
- Diagramatic representation of Volcanic cones:



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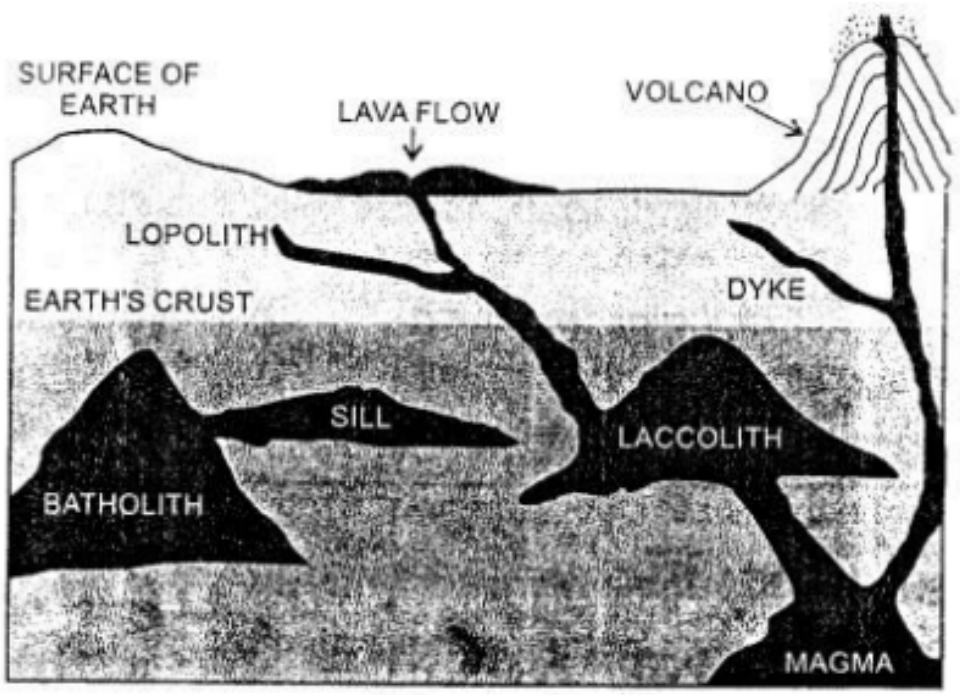
- *<u>Shield volcano:</u>*
- It is formed from eruptions of very fluid basaltic magma.
- It is formed along hotspots due to mantle plumes.
- They are the largest of all volcanoes.
- They are not very steep.
- Examples Mauna Leo and Mauna Kea.
- *<u>Ash-cinder volcano:</u>*
- They are formed due to the accumulation of loose particles around the vent.
- They are dominated by ash, debris, and other volcanic material with less lava.
- An example includes the Taal volcano in the Philippines.
- *<u>Composite volcano:</u>*
- They are formed due to the accumulation of viscous lava which is of lower fluidity.
- The alternating layers of ash and lava provide for composite structure.
- They are one of the steepest of all types of volcanoes.
- For Example Mount Fuji in Japan, Krakatoa, etc.
- ***Crater:***
- It is a funnel-shaped depression formed at the mouth of the volcano.
- The crater filled with water results in the formation of a crater lake.
- ***Caldera:***

- It is a greatly enlarged depression resulting from the violent eruption volcano resulting in the subsidence of much of the Volcano.
- When water is filled it results in the caldera lake.
- ***Flood basalt province:***
- When a highly fluid basaltic magma erupts over a plateau region it flows for a long distance covering large areas and creating a flood basalt province
- For example, the Deccan province and the Columbian plateau.
- ***Hot springs/thermal springs:***
- When water sinks deep enough beneath the surface to be heated by a hot spot or a magma chamber or a geothermal belt it rises to the surface continuously without any explosion.
- Water from hot springs contains dissolved minerals with medicinal value.
- For example, the hot springs in USA, Iceland, and India (Rajgir, Manikaran, Manali, etc).
- ***Geysers:***
- They are the fountains of hot water and superheated steam, ejected at regular intervals with explosions.
- For example Geysers of Iceland, New Zealand, USA(Old faithful Geysers)
- ***Fumarole:***
- It involved continuous jet-like emission of steam and other gases.
- For example Fumarole in Iceland.
- ***Mud volcano:***
- It involves the eruption of mud mixed with water and gases.
- They may not contain magma.
- For example Baratang island of Andaman and Nicobar.

Intrusive features of Volcanism(6.19 PM):

- These are the landforms that are formed inside the earth's surface as manga cools down below the surface.
- *Diagrammatic representation of Extrusive volcanic landforms:*



Intrusive Landforms

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- *Batholiths:*
- It is a long irregular dome-shaped structure.
- They are large bodies formed by the cooling of magma along deeper layers of the lithosphere.
- *Laccoliths:*
- It is a mushroom-shaped feature formed due to the intrusion of magma.
- It maintains contact with the magma chamber.
- *Lopoliths:*
- When the magma solidifies along a saucer-shaped depression it results in lopoliths.
- *Sill:*
- These are formed due to the cooling of magma along horizontal bedding planes.
- *Dykes:*
- These are vertical wall-like structures formed by the cooling of magma.

Geomagnetism(6.49 PM):

- The magnetic field associated with the earth is called geomagnetism.
- Earth has an outer rocky mantle, below which there is the liquid outer core that surrounds the solid inner core.
- It is considered that the motion of iron charges in the liquid parts of the earth's core generates a Magnetic field.
- *The motion of iron-charged particles is caused:*
- Rotation of earth.
- A convection current of molten material is generated in the outer core due to heat released from the earth's inner core.
- The motion of ions in this molten material produces an electric current through the earth's core that results in a magnetic field around the earth called geomagnetism.
- *Palaeomagnetism and Polar wandering:*
- It is the historical study of earth magnetism through rock called Paleomagnetism.

- When the ingenious manga crystallizes, the crystal of ferromagnetic minerals such as iron, titanium, etc acquires stable magnetism which becomes frozen as the magma cools down.
- This acquired magnetism is called fossil magnetism.
- Such rocks will have ferromagnetic minerals aligned in the same direction as that of the geomagnetic field at the time of its consolidation.
- When paleomagnetic pieces of evidence for a given region were checked over a long period a gradual change in direction was detected.
- It shows that the position of magnetic poles has moved over of period, this movement is called polar wandering.
- During the study of the oceanic surface, some of the rocks were found pointing toward the north and some toward the south.
- It is inferred that the rocks can not change the polarity but it is the earth's magnetic field that reveres its polarity.
- Such a reversal of magnetic field can be brought by the reversal of convection currents in the outer core.
- The magnetic reversal occurs every 2,00,000-300,000 years.
- *Aurora:*
- The magnetosphere of the earth's atmosphere blocks the charged particles from solar winds.
- But some of these charged particles get trapped near polar regions where the magnetic lines are straight.
- Due to the interactions of charged particles in the earth's atmosphere near polar regions, colourful lights are released called aurora.
- It is called Aurora Borealis in the northern region and Aurora Australis in the southern region.

Earthquakes(7.52 PM):

- The vibration or oscillation felt near to the earth's surface due to transient disturbance of the elastic or gravitational equilibrium of the rocks at or beneath the surface of the earth.
- *The various causes of earthquakes are:*
- Plate movements.
- Minning.
- Reservoir-induced seismicity(RIS).

The topic of the next class: Earthquake to continue.

Geography Class 16

Last class revision (5:16 PM).

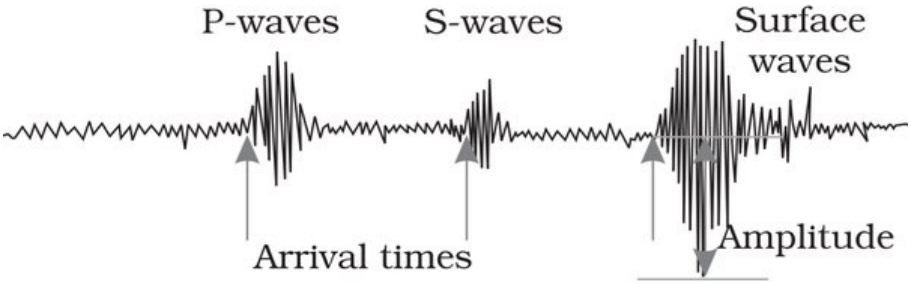
Earthquakes (continued) (5:28 PM):

- ***Causes of earthquake:***
- *Natural causes:*
- Plate movements
- Volcanic eruption.
- Landsides.
- Land Subsidence.
- Meteoritic impact.
- *Anthropogenic causes:*
- Mining and blasting.

- Nuclear explosion.
- Reservoir-induced seismicity(RIS).
- The scientific study of earthquakes is called seismology.

Earthquake waves(5.35 PM):

- Diagrammatic representations of Earthquake waves:



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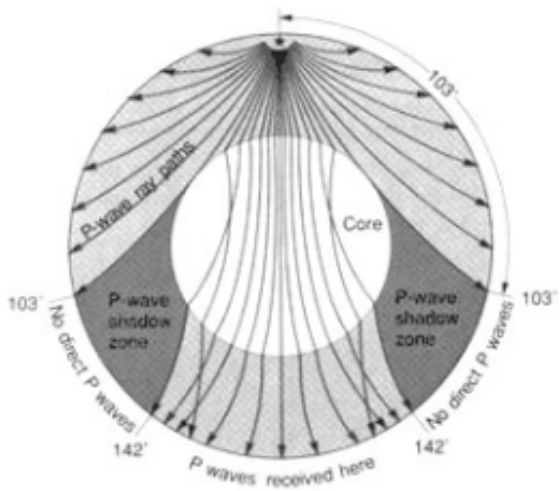
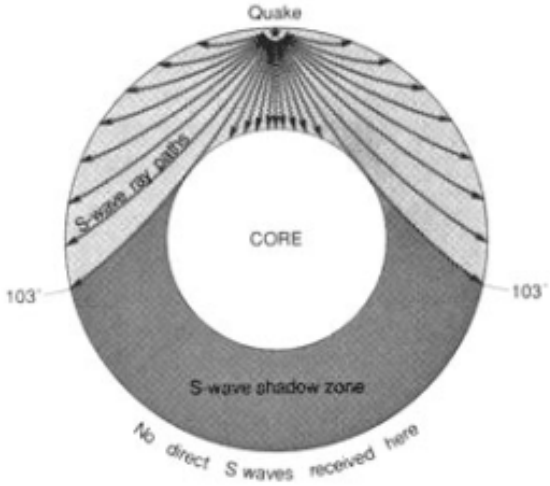
- During an earthquake, the point where the energy is released below the surface is called the focus/hypocentre.
- The point on the surface nearest to the focus is called the epicentre.
- An earthquake generates two types of waves, body and surface waves.
- ***Body waves:***
- These earthquake waves are generated at the focus and move through the earth's interior.
- There are two types of body waves, P-wave and S-wave.
- *P-waves(Primary waves):*
- These are the first to be detected during an earthquake.
- They are also called compressional waves and longitudinal waves.
- They oscillate back and forth in the direction of propagation.
- P-Waves depend on the density and rigidity of the material.
- P- waves can travel through solid, liquid, and gas.
- *S-waves(secondary waves):*
- S waves oscillate perpendicularly to the direction of propagation.
- They are also called transverse waves.
- The S-wave movement depends on the material's density and rigidity.
- Therefore S waves cannot pass through liquid and gas.
- ***Surface waves:***
- These are the earthquake waves confined to the surface of the earth.
- They are generated due to the interaction of body waves with the surface and generally at the epicentre.
- Surface waves are much slower than body waves but cause more damage.
- Examples are Love waves and Raleigh waves.

Shadow zones(6.18 PM):

- Diagrammatic representation of shadow zones:





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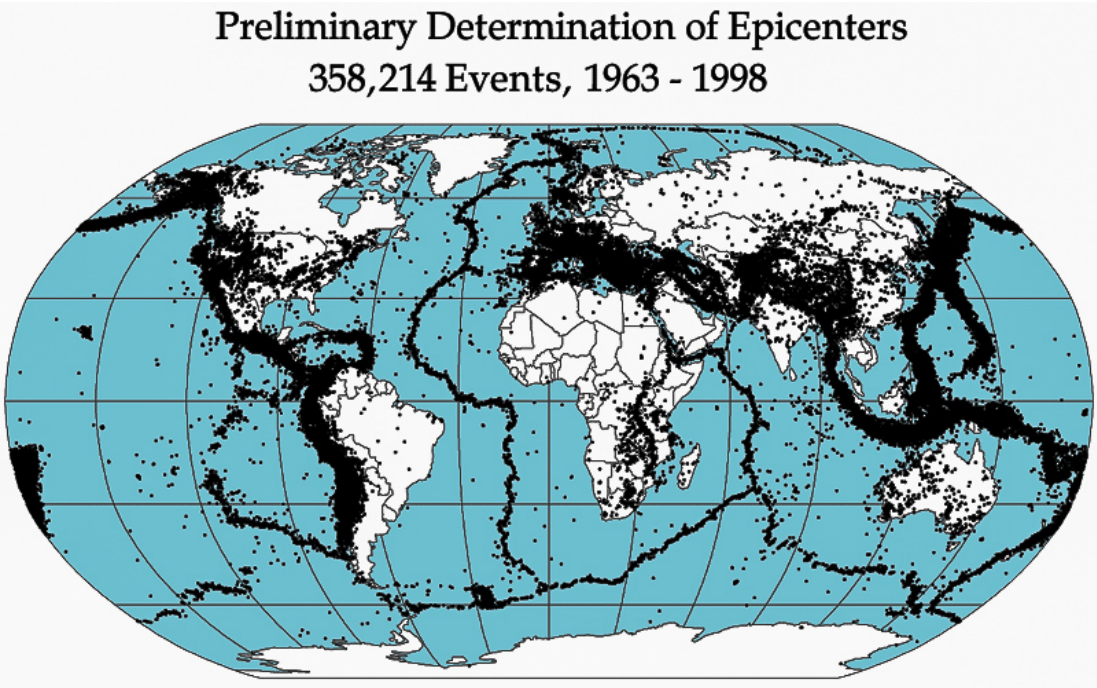
- The seismic waves undergo reflection, bending, or changes in direction while passing from one medium to another.
- This results in the creation of shadow zones.
- The Shadow zone is the zone in the earth's interior from where earthquake waves are not reported.
- *P-wave shadow zone:*
- P-waves are not detected between 105-145 degrees from the focus resulting in the P-wave shadow zone.
- It is due to a change in direction of P-waves as it enters the outer core.
- *S-wave shadow zone:*
- S-waves cannot pass through the liquid outer core.
- Therefore beyond 105 degrees, we observe S-wave shadow zones.
- By studying the different earthquake waves, the extent of their shadow zone, and other characteristics we can conclude about the earth's interior.

Type of earthquake (6:32 PM)

- Shallow focus earthquake occurs at the depth of 0-70 km.
- Intermediate focus earthquake occurs at the depth of 70-350 km.
- Deep-focus earthquake occurs at the depth of more than 350 km.

Distribution of earthquakes (6:35 PM):

- ***Circum-pacific belt:***
- Around the pacific ocean.
- Along the west due to ocean-ocean convergence.
- Along the east due to ocean-continent convergence.
- ***Mid-oceanic ridges:***
- Around the regions of ocean-ocean divergence
- ***Meditarian and trans-asiatic belt:***
- Continent-continent convergence.
- ***East African rift valley:***
- Continent-continent divergence.
- Diagramatic representation of the distribution of earthquakes:

- 

Measurement of earthquakes (7:03 PM)

- The earthquake waves can be recorded by a seismograph.
- The record produced by the seismograph is called a seismogram.
- ***Magnitude and intensity of the earthquakes:***
-

Magnitude:	**Intensity:**
It is the measure of the amount of energy released during an earthquake.	Intensity is the assessment of visible damage caused by an earthquake.
The magnitude remains constant for earthquakes.	Intensity decreases away from the epicentre.
It is an objective measure.	It is a subjective measure.

The magnitude is measured by the Richter scale. There is no upper limit for the Richter scale. Richter scale is a logarithm scale that is 1 unit change means a 10 times stronger earthquake.	Intensity is measured by a Modified Mercalli scale. The range of the Mercalli scale is 1-12.

- The highest recorded earthquake is a 9.5 Richter scale in Chile in 1960.
- 2004 Indian ocean(Indonesia) earthquake was 9.1 magnitudes.
- 2011 japan earthquake was 9.1 magnitudes.
- ***Earthquake in India:***
- 2001 Bhuj earthquake was 7.7 magnitudes.
- The 2005 earthquake in Kasmir was of 7.6 magnitudes.
- 2011 Sikkim-Nepal earthquake was of 6.9 magnitudes.
- The 1993 Latur earthquake was of 6.2 magnitudes.
- 1991 Uttarkashi earthquake was of 6.1 magnitudes.

T-sunami (7:31 PM)

- A tsunami is a Japanese word for harbour waves.
- They are also called seismic sea waves.
- They are four stages in the occurrence of a tsunami:
- ***Generation:***
- Tsunamis are generated due to vertical displacement of the column of seawater which causes by:
- Earthquakes, normal and reverse fault earthquakes along the oceanic floor.
- Underwater Volcanic eruptions.
- Underwater landslide and subsidence.
- Meteorites impact.
- Underwater nuclear test.
- ***Propagation in deep water:***
- Along deeper waters, the speed of waves is very high and the amplitude will is low.
- Therefore tsunamis will not be visible in the open ocean nor will cause any damage.
- ***Propagation in shallow water:***
- As depth decreases, the amplitude increases, and the speed of waves decreases.
- This causes a pilling up of water near the coast, generating very high waves, this is called the shoaling effect.
- ***Landfall:***
- Where the waves break and submerge the coastal regions.

The topic of the next class: Exogenetic movements.

Geography Class 17

Last class revision. (5:11 PM)

Exogenic Movements: (5:18 PM)

- These are the earth's movements caused due to various forces acting from outside the earth's surface.
- The source of energy for these movements are sunlight and gravity.
- Different agents of exogenic movements are wind, river, groundwater, waves, and glaciers.
- Exogenic processes involve aggradation and degradation.
- Aggradation is the action of various processes that cause deposition on the surface to bring uniformity of grade i.e. slope.
- Degradation or denudation involves the combined action of various processes that cause the wearing away of the earth's surface and results in general lowering and levelling out of the earth's surface.
- So, degradation involves weathering, erosion, and mass movements.

Weathering (5:41 PM):

- It is the total effect of various processes that cooperate in bringing about the decay and disintegration of rocks involving no large-scale transportation.
- ***Factors affecting weathering:***
- *1. Rock type and structure:*
- It includes characteristics like origin, massiveness, porosity, permeability, presence of fractures, etc.
- For example, sedimentary rocks are weathered faster than igneous rocks.
- *2. Slope and aspects:*
- Steep slopes will greatly add weathering as disintegrated material gets removed faster.
- The slopes which are exposed to wind, rainfall, and sunlight are more prone to weathering.
- E.g. south-facing slopes of the Himalayas.
- *3. Temperature:*
- The range of temperature, particularly the diurnal range subjects the surface layer to expansion and contraction continuously causing a physical breakdown.
- Temperature also increases the rate of certain chemical reactions.
- *4. Water:*
- When water is added or removed, it imposes mechanical stress on the rocks and causes them to split.
- Water also decides the rate of chemical reactions.
- *5. Biological agents:*
- These include the action of plant roots, micro-organisms, and burrowing animals.

***Types of* weathering (6:15 PM)**

- ***Difference between Mechanical and Chemical weathering:***
-

Mechanical (Physical) Weathering	*Chemical Weathering*
The disintegration of rocks takes place without any change in the	The disintegration of rocks takes place with changes in the

chemical constituents of the rocks	chemical constituents of the rocks
Factors such as temperature, moisture, frost action, and wind cause the physical breakup of the rocks.	Temperature, moisture, etc. cause minerals in the rocks to either dissolve in water or change their composition
It is more rapid in desert climates	It occurs in moist and cool climates
Rocks are effect to great depth	It mostly takes place near the surface of the earth
Its agents are temperature and moisture	Its agents are water, oxygen, carbon, and various organic acids

- ***Physical weathering:***
- *Crystal growth*
- *Growth of water crystal*: When water freezes it expands and finds its way into cracks or pores, it wedges apart rock fissures and joints. In desert regions, it results in frost shattering, in glacial regions it causes freeze-thaw.
- *Salt crystal:* Growth of salt crystal by crystallization as moisture evaporates creating pressure in pore spaces and cracks. It is also called Haloclasty (salt weathering).
- *Pressure or thermal expansion:*
- *1. Exfoliation*: Rapid expansion and contraction of surface rocks weaken the outer shell creating cracks and causing the rock shell to peel off
- *2. Pressure release:* Confining pressed from the weight of overlying rocks is released, when the upper layer of rocks is removed by degradation. This causes rocks to expand leading to the development of fractures.
- *3. Block disintegration:* Here weathering causes the breakdown of rocks in blocks.
- *4. Granular disintegration:* Here breakdown occurs in granules.
- ***Chemical Weathering:***
- *Hydrolysis:*
- It is the chemical union of water and minerals.
- It produces solutions.
- *Hydration:*
- It is when minerals incorporate water into their molecular structure.
- It causes swelling.
- *Oxidation and Reduction:*
- It involves the addition or removal of oxygen.
- It causes discolouration. E.g. rusting of iron.
- They are the first visible sign of chemical weathering.

- *Carbonation:*
- Rainwater absorbs carbon dioxide and gets converted into carbonic acid.
- Limestone and dolomites are altered by the dissolving action of carbonic acid.
- ***Biological Weathering***
- It is a type of weathering due to the action of biological agents.

Erosion: (7:20 PM)

- Erosion refers to the wearing away of the land surface by the mechanical action of debris derived from weathering as they are transported to a different location by agents of wind, water, glacier, etc.
- ***Types of Erosion:***
- *Abrasion:*
- Wearing away surfaces by mechanical processes such as rubbing, scratching, polishing, etc.
- Abrasion in water is called *corrasion.*
- *Attrition:*
- It involves the reduction in the size of fragments by friction and impact during transportation.
- It breaks down the load into finer sediments.
- *Cavitation:*
- The collapse of water bubbles in rivers leads to an explosion sending a shock wave that tends to disintegrate the adjacent rocks which are cavitation.
- *Corrosion:*
- It involves the breaking of rocks by the solvent and the chemical action of water.
- *Deflation:*
- The lifting and removal of dust and sand by wind are called deflation.
- *Hydraulic action:*
- Breaking of rocks by fast-moving water.
- *Plucking:*
- As the glacier moves, they drag the rocks and disintegrate them into smaller pieces.

Mass Movement: (7:53 PM)

- The weathered material moving downhill slopes under the influence of gravity with or without the assistance of running water is called a mass movement.
- It is a large-scale breakdown or disintegration.
- It is also called mass wasting.

Topics for the next class: Types of mass movements, and different types of landforms.

Geography Class 18

A brief review of the previous class and doubts. (5:08 PM)

Types of Mass Movements.

- **Slow**:
- Creep: The slow downhill movement of the soil and weathered mantle along the mountain slopes.
- It is a continuous movement but difficult to notice
- Solifluction: It involves the slow movement of soil mixed with water as a viscous fluid
- It is more active in glacial areas
- **Rapid:**

- Landslide: A landslide is the movement of a cross-section of a hill slope under the influence of gravity is called as landslide
- The material maintains continuous contact with the surface as it moves
- Avalance: Similar to landslides, is sudden collapse and sliding of snow-covered regions
- Earthflow: Thye occur along terraces and hillsides that are capable of flowing as a single unit when saturated with water below the top layer
- Mudflow: It involves the rapid movement of masses of uncompacted soil ie suddenly saturated with water
- Rock slide/fall: It is a rapid movement including rock material either sliding along the mountain slopes (rock slide) or falling vertically

Landforms: (6:04 PM)

- **Agent and its topography:**
- River - Fluvial Topography
- Ground - Karst topography
- Waves - marine Topography
- Wind - Desert or Arid Topography
- Glacier - Glacial topography
- Each causes erosional as well as depositional topography

Riverine Topography:

- It is a topography which is built by riverine erosional and deposition
- 3 stages in riverine topography: Youth Stage, Mature Stage, and Old Stage
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Index	**Youth Stage**	**Mature Stage**	**Old Stage**
Slope	High	Moderate	Low
Energy	High	Moderate	Low
Erosion	High - Vertical	Moderate Lateral	Low or No erosion
Deposition	Very low	Moderate	High
Meandering	No	Moderate	High

Features of Riverine Topography: (6:52 PM)

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Topography	Processes	Erosional	Depositional
▪ Riverine (Agent - River water)	▪ Corrasion ▪ Attrition ▪ Cavitation ▪ Corrosion	▪ V-Shaped Valleys ▪ Gorge and Canyon ▪ Waterfall,	▪ Slip of slope ▪ Levee ▪ Flood Plain ▪ Ox-Bow lakes

	▪ Hydraulic ▪ V-Shaped Valleys	Cataract, Rapids ▪ Plunge pools and Pot Holes ▪ River Capture and wind Gap ▪ Meander - River Cliff	▪ Alluvials fans and Delta ▪ Estuary

<u>Topics for the next class</u>: Continuation of Landforms, etc

Geography Class 19

Geography Class 19 [17:10:00]
A brief overview of the previous class i.e. Landforms of riverine topography
Karst Topography/ Limestone topography [17:25:00]

- [* If Fresh water is 100 units then 67-68 units are ice caps, 30 units are groundwater, 1-1.5 units are water vapor, and 0.3 units are water in rivers, lakes, etc]
- Underground Rock systems are called Aquifers
- When water (Containing carbonic acid) falls on the surface, It melts the rocks very easily. It will sip in and joins the groundwater

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- **This term was used in Eastern Europe, erstwhile Yugoslavia.**
- Many large Caves are found- in erstwhile Yugoslavia, countries of Southeast Asia[Thailand, malyasia], Meghalaya, Mexico - Yucatan peninsula [The meteorite fell here led to the extinction of the dinosaurs, Mayan civilization flourished here, devoid of surface Fresh water, they survived through the Karst topography], Western Australia, In India these are found in Meghalaya
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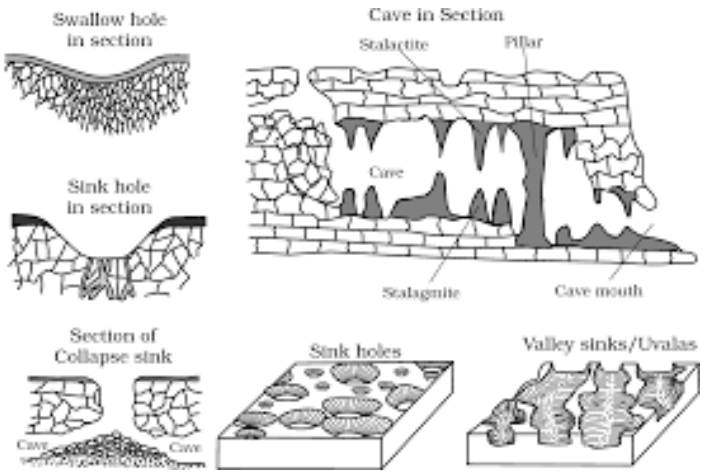
Topography	Processes	Erosional features	Depositional features

Karst/ Groundwater	**Conditions that led to this karst topography** ○ Soft and Permeable rocks in the upper layer of lithosphere such as limestone, dolomite ○ Moderate to heavy precipitation ○ Hard strata of rocks at the sub-surface level **Processes** ○ Corrasion ○ Corrosion ○ Attrition ○ Carbonation ○ Hydrolysis [Less hydraulic action, No or less attrition]	**Features** ○ **Sinkholes** are created- very wide and huge depression ○ **Swallow hole-** Shallow version of sinkholes ○ **Doline-** Sink holes merge together to form Doline ○ **Uvala-** Dolines merge together to form Uvala [Refer to the below diagram] ○ **Blind valley/karst valley-** The river disappears and only valleys are visible, the rivers flow underground. ○ **Caves-** Ground water creates caves Example- Mexican caves, Meghalaya caves	**Features** ○ **Stalactites-** These develop from the roof from the dripping water ○ **Stalagmites -** The deposits develop upward from the floor. ○ **Cave Pillar-** When both the stalactites and stalagmites meet. [Important activity- Cave diving]
Regions	Erstwhile Yugoslavia, South-East Asia, Meghalaya in NE India, Mexico, Western Australia		

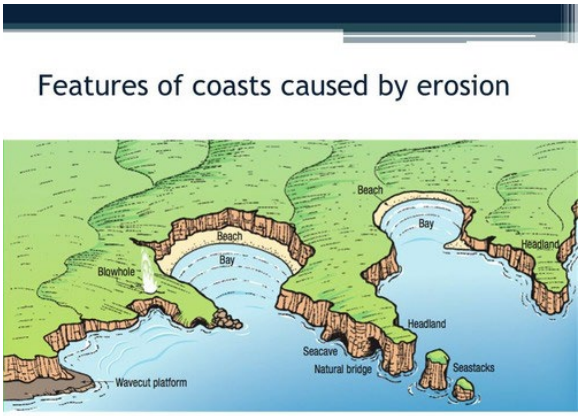


Marine Topography [18:15:00]

- Agent- Waves
- Source of energy- WInd (Driven by sunlight due to differential heating)
- Erosion and deposition are dependent on the type of coast
- If forward force is more then it will lead to erosional features whereas if backward force is more then it will lead to Depositional features
- On the eastern coast deposition happens and on the Western coast erosion happens (The exception is the Kerala coast)
- **Processes-** Corrasion, Attrition, Cavitation, Corrosion, Hydraulic action (No deflation, no plucking)

Erosional features

- **Cliffs-** Near to coastal region [due to erosion-steep slope]
- **Wave cut platforms**- Upper part is eroded but the lower part is not eroded [The lower part is hard rock], a platform is left due to differential erosion
- **Bay-** Too much erosion on the inner side, and wave action is more. And it creates a semi-circle shape [Differential erosion] {* **Gulf** is not a part of marine topography, it is a waterbody+ Gulf have narrow mouth or elongated mouth. For example- the Persian gulf, and Mexican Gulf}. When the Mouth of the Gulf is too wide then it is called a Bay. Example- the Bay of Bengal, Hudson bay.
- **Hanging valley** - It is a river valley that ends abruptly into the coastal region. That river valley is hanging into the ocean. For Example- Found in Norway.
- **Sea caves-** Due to continuous waves hitting the cliffs, creates caves from the seaside
- **Sea arch or natural bridge-** Development of sea caves from both sides, Gradually it will deepen and merge together. When two sea caves merge together it is called a sea arch.
- **Sea stack-** When the roof of an arch is broken by erosion or under its own weight or due to any other reason a part of the original rock remains standing as a solitary mass. It may be the rock forming the side of the arch. This type of feature is called a sea stack.
- **Sea stump-** The smaller version of the sea stack is called the sea stump
- **Blow holes-** When a powerful wave hits, it creates a hole in the upper part. Then the water comes out from the blow holes



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Depositional features

- **Wave-built platform**- When the sea recedes and due to the deposition of sand a platform can be built, for Example- near the bay
- **Beach**- Where the wave speed reduces and the deposition of sand is more. In India, along the eastern coast, we have more no. of beaches [The longest is Marina beach, Chennai]
- **Off-shore sand bars**-When the slope is less and the water level is also less, then it creates the feature of sand bars. Sand bars not connected to the land are called offshore sand bars [Mostly found on the East coast, one of the sand bars is strategically located in **Sriharikota** (ISRO satellite center, on Pulicat lake)]
- **Sand spit**- When the sand bar is connected on one end with land
- **Tombolo**- When the sand bars are connected on two sides with the land.
- **Lagoons**- A lake near the coastal region is called a lagoon. It is separated from the main part. [* Kerala state is known for the backwaters, houseboats, etc]
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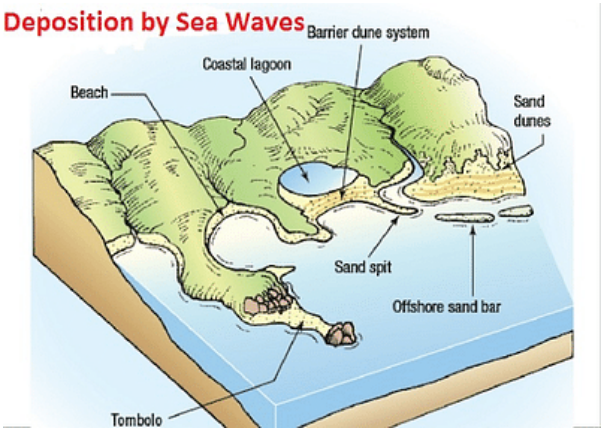
- 

Arid/ Desert topography [19:10:00]

- **Agent** - Wind [It is more of physical weathering due to sunlight], Erosion [Wind is at very high speed, as there is no vegetation]
- **Erosional Process involved**- Attrition, Abrasion, Deflation (Lifting action of wind) [No corrosion, cavitation as they require water], No plucking
- [* Reasons for Desert formation- Low precipitation less than 25 cm]

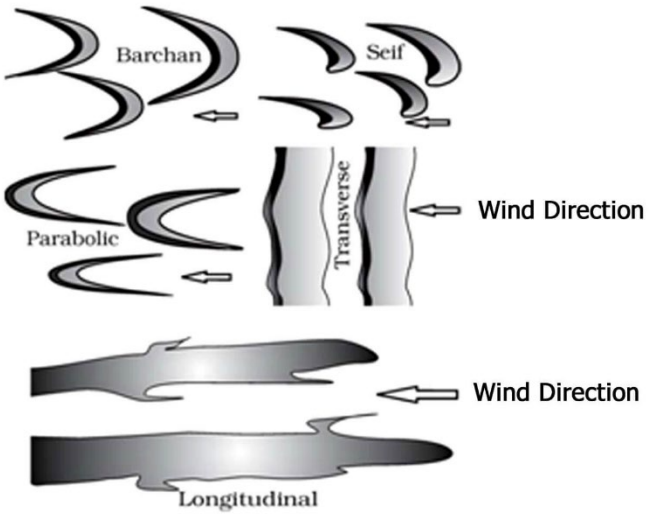
Erosional features

- **Deflation hollows**- The hollows or empty spaces created by the winds. These are created by **Deflation**
- **Ventifacts**- Abrasion causes the polishing of the surfaces, and the polished surfaces are called ventifacts

- **Mushroom Rocks-** Rocks near the surface go faster weathering than the above one [Wind action is faster, as sands carried by the wind are below], this creates the mushroom rocks. (Caused by Abrasion near the surface)
- **Zeugen**- Continuous layer of hard and soft rocks in a **Vertical direction**, Soft rocks will easily be eroded. Differential erosion will lead to Zeugen. It is caused by abrasion
- **Yardungs-** When differential rocks are there in a **horizontal position.**
- **Inselbergs-** Remanant of hills. In the central part, the hard rocks are there which take a longer time to erode.

Depositional features/ landforms

- Sands are more in the desert because of attrition
- **Sand dunes**- famous beaches- Thar desert [**Marusthali**], Ladakh [Nubra valley- **Hundar sand dunes**], Arabian sand dunes [**Rub-al-khali** sand dunes], Namib desert [The largest sand dunes are in Namib desert]
- **Barchans**- crescent-shaped sand dunes are called barchans. If the limbs are in the direction of the wind
- **Parabolic-shaped sand dunes**- if limbs are opposite to the direction of the wind
- **Seif-** Barchan with one limb
- **Transverse sand dunes**- Perpendicular to the direction of the wind
- **Longitudional sand dunes**- Parallel to the direction of the wind
- **Bolson lake/Playas lake-** Refer to the diagram [Above]
- **Oasis-** Fertile land around the playas region
- **Pede-plain-** Pediplains are formed when high relief features in deserts are lowered to low featureless plains due to wind activity. The convergence of pediments forms a Pede-plain, which is a largely flat landscape

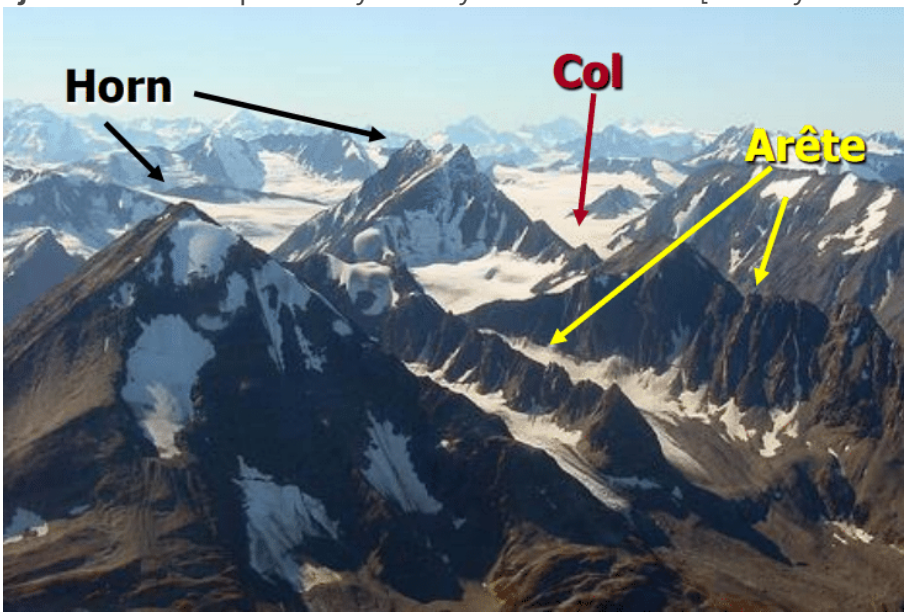


Glacial Topography [19:39:00]

- Formed along the mountain sides,
- **During winter**- when snow is deposited on the mountain sides, a thin layer of snow is created. Where there is more depression, more accumulation of snow occurs. Water will sip into the gaps of the snowflakes and it will get harden. Through continuous melting and freezing, it becomes a thick block of ice
- **During summer**- The entire block of ice will start to melt from its base. It will start to move. It has a very high erosional capacity
- **Agent**- Glaciers [Capable of movement, largest force]

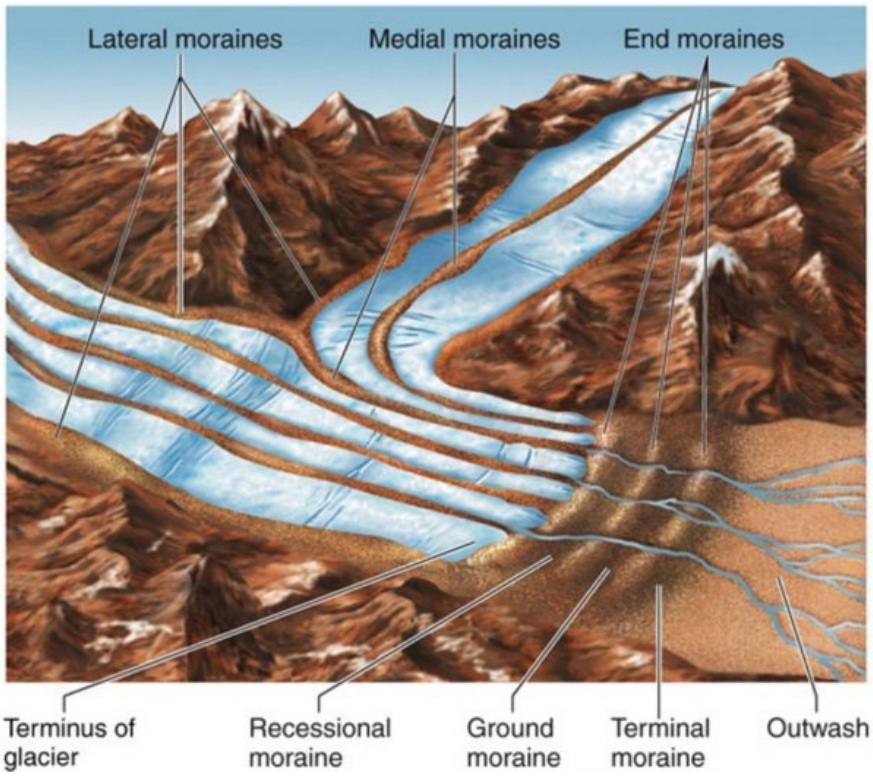
- **Process involved-** Abrasion, plucking, erosion
- [* 5 Big lakes are the result of glacial erosion]

Erosional features

- **Nivation hollows**- Small depressions where the glaciers start to build up [from mountain size to huge glacier]
- **Cirque**- Continuous circulation of the glacier, enlarged depression
- **Arete**- It separates two Cirques
- **Horn**- When cirques are developed from all sides, and the top mountain becomes too narrow and steep. For example- **Matterhorn** [Alps mountain], Kailash mountain, Mt. Everest
- **Col-** When the horn is entirely eroded and the upper part is removed. **Indira col** [North of siachen]
- **U-shaped valley**- It is the valley created by the glaciers. When multiple glaciers merge together.
- **Hanging valley-** Small U-shaped valley merging with the big U-shaped valley.
- **Fjords-** The U-shaped valley directly enters the ocean [Norway is known for Fjords]
- 

Depositional features

- **Moraines-** Unstratitifed mix of sand and ice carried by a glacier- lateral, medium, end moraines, and at the base is called Ground moraines
- **Moraine capped lakes-** When these lakes outburst it is called GLOF [Glacier lake outburst flood- example- **Chorabari lake**- north of Kedarnath]
- **Tillites-** It is a heterogeneous mixture of Clay and Till from pre-historic times
- **Drumlins**- Semicircle depositions are called baskets of egg topography or Drumlins
- **Esker-** When it is forming a longitudinal ridge it is called Esker



- **The Topic for the next class-** Climatology.

Geography Class 20

Questions on topics taught in the previous class (05:12 PM)

Climatology (05:21 PM)

- **Atmosphere:** The gaseous envelope which **surrounds the earth** from all sides and is attached to the earth's surface by gravitational force.
- **Weather:** It is the day-to-day state of the atmosphere.
- **Climate:** The average atmospheric conditions of an area over a considerable period of time calculated for a **minimum of 30 years**.

Composition of Atmosphere (05:45 PM)

Table 8.1 : Permanent Gases of the Atmosphere

Constituent	*Formula*	*Percentage by Volume*
Nitrogen	N_2	78.08
Oxygen	O_2	20.95
Argon	Ar	0.93
Carbon dioxide	CO_2	0.036
Neon	Ne	0.002
Helium	He	0.0005
Krypto	Kr	0.001
Xenon	Xe	0.00009
Hydrogen	H_2	0.00005